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Project Title:	Functional Genomic and Signaling Responses to HZE Particle Radiation		
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Contact Monitor:		**Contact Phone:**	
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Task Description:	Recent advances in gene expression profiling have broadened our understanding of the molecular mechanisms of response to a wide range of environmental insults, including toxic chemicals and radiation. Such studies hold great promise for the development of exposure and diagnostic biomarkers, as well as pointing the way to intervention strategies through mechanistic understanding. The majority of such studies, however, have considered only low linear energy transfer (LET) radiation, and it is unknown to what extent this knowledge may be applicable to the high LET heavy ions prevalent in the space radiation environment. This proposal seeks to apply functional genomic analysis in conjunction with the irradiation capabilities of the NASA Space Radiation Laboratory (NSRL) to address questions of the molecular responses to highly-charged high-energy heavy ion (HZE particle) irradiation. A more comprehensive understanding of the molecular signaling pathways activated in response to damage by HZE particles will identify biomarkers of exposure and damage, and molecular targets for modification of the radiation response, as well as enhancing our understanding of the mechanisms by which HZE particle exposure may lead to the development of cancer and other late effects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Funding was provided for one year in order to obtain preliminary data with heavy ions and to demonstrate feasibility of the proposed studies. We were able to obtain beam-time and complete three independent irradiations with 1 GeV 56Fe ions at NSRL during the two runs (NSRL-5 and NSRL-6) that were available during the funding period. While microarray analysis is ongoing, the data so far demonstrate clearly that the experiments were successful. This data and will provide a solid basis for shaping future studies. Survival While culturing cells in the presence of activated medium did not significantly affect cloning efficiency ($p>0.6$), the lowest dose used (10 cGy) resulted in a nearly 20% decrease in cell survival ($p<0.05$). The overall TK6 survival curve showed a linear decrease in survival with increasing dose, with overall survival levels in agreement with what had been predicted from the literature. The NH32 cells were treated with only two doses of iron ions, so the shape of the survival curve could not be accurately determined. Survival of the p53 knock-out cells was similar to that of the parental TK6, with perhaps a slight enhancement at the higher dose. Apoptosis Nuclear morphology was scored in DAPI-stained cells as a measure of apoptosis at 3 hours after irradiation, and at 24-hour intervals for the first week after exposure. The activated medium did not produce any significant deviations from the level of apoptosis in the unirradiated controls. The lowest (10 cGy) dose produced a slight elevation of apoptosis, but this was only significant at 2 days post-irradiation ($p<0.05$). Induction of apoptosis increased with increasing dose, but in all cases had returned to near-baseline levels by one week after exposure. No further elevation in apoptosis was observed up to 3 weeks post-treatment. Interestingly, the p53 knock-out cells showed nearly the same peak levels of apoptosis as the p53 wild-type parent cell line, but with different kinetics of expression. Apoptotic morphology took longer to manifest in the absence of wild-type p53 protein, and apoptotic cells persisted in culture somewhat longer. Cell cycle distribution and mitotic index Culture in the presence of activated medium did not alter cell cycle distribution. No significant alterations in the mitotic index ($p=0.8$) were seen in cells cultured with activated medium. In contrast, the lowest dose, 10 cGy, did induce a transient cell cycle disturbance, quantified as a decrease in mitotic index. This decrease was significant at 3 and 8 hours after treatment, with the cultures recovering normal levels of mitosis by 24 hours. Increasing doses resulted in more profound and more prolonged loss of cells from M-phase. Doses of 167 cGy or above essentially eliminated the presence of mitotic cells. Although the initial effects were virtually identical with or without p53, the knock-out cell line appeared to resume normal cell cycle slightly more rapidly than the wild-type TK6. All cultures had recovered normal levels of mitosis by one week post-exposure, and further disturbances were not noted two or three weeks later. Expression of CDKN1A Expression of the CDKN1A gene was measured at multiple time-points. Incubation of cells in activated medium did not result in any significant change in CDKN1A expression at any time-point measured ($p=0.85$). Peak expression in TK6 cells exposed to 56Fe ions occurred between 3 and 4 hours post-irradiation, consistent with low LET experiments. However, in contrast to the rapid return to near normal gene expression levels expected with low LET, 56Fe ion exposure yielded a prolonged elevation of CDKN1A expression, with little decrease from maximum levels during the first 24 hours. By one week after treatment, levels had returned to normal, and did not increase further through week three. This protracted elevation was observed even at the lowest dose used, which significantly elevated CDKN1A expression throughout the first 24 hours ($p<0.0001$). The dose response remained fairly linear at all times between 2 and 24 hours. The dose-response is illustrated for TK6 at the peak induction time of 3 hours. There is a slight flattening of the dose-response curve at the highest dose, as is typically seen with highly toxic treatments. Interestingly, there is also a discontinuity evident at the low end of the dose-response curve. This mirrors the response we have seen previously with low LET, where gene induction responses for several genes were found to be linear within a certain dose range, but did not extrapolate back to baseline levels at zero dose. The ionizing radiation response of CDKN1A is dependent on TP53, and as expected, no induction was observed at early times in NH32, the p53 knock-out cell line. At 24 hours, there was a slight elevation in CDKN1A levels in the 56Fe ion treated NH32 cultures, however. This was significant at the 167 cGy dose, although the magnitude of induction remained far below that observed in TK6, the TP53 wild-type cell line. Preliminary microarray results Because the NSRL-6 beam-time, which provided the majority of our samples and data, did not occur until the very end of the one-year funding period, the microarray analysis is still ongoing. This data represents the heart of this project, and will provide the most insight for the direction of future proposed studies. In an initial comparison of early gene expression changes by microarray in TK6 cells treated with 56Fe ions, neutrons, and g-rays, both similarities and differences in the early response to HZE and low LET irradiation can be seen. As additional data is added to this analysis, a clearer picture of different response pathways should emerge. Careful analysis

	of dose and time dependence will be needed to determine if the differences observed represent some unique responses, or alterations in the time or dose responsiveness of genes commonly regulated by low LET. Genes responding to 56Fe ions and g-rays at 3-4 hours after treatment were annotated using EASE and compared to uncover commonly or differentially involved response pathways. Among down-regulated genes, there was a significant involvement of mitosis regulating genes in both the 56Fe ion- and g-ray-treated cells. Categories of up-regulated genes were more diverse, however. Although a slight over-representation of cell cycle regulatory genes was evident in the response to both radiation qualities, the major functional categories involved in the response to 56Fe ions were mitochondrial and other extra-nuclear genes. These same categories of genes were not significantly responsive to g-rays, indicating that different response pathways may indeed be activated by different radiation qualities. It should also be noted that although the initial analysis is focusing on the first 24 hours of response, high quality RNA and supporting cell cycle and apoptosis measurements were obtained at one, two and three weeks after exposure to 56Fe ions. This will enable the extension of these studies to later time-points, beyond the range of most previous studies, even those using only low LET.
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